

No. of Printed Pages : 4  
Roll No. ....

220952

**5th Sem/ Electrical**  
**Subject : Power System**

Time : 3 Hrs.

M.M. : 60

**SECTION-A**

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Chances of occurrence of corona are maximum during:
- a) Humid weather      b) Dry weather
  - c) Winter              d) Hot summer
- Q.2 The Distributors in residential areas are
- a) Single phase two wire
  - b) Two phase four wire
  - c) Three phase three wire
  - d) Three phase four wire
- Q.3 The sag of transmission line is least affected owing to
- a) Weight of the conductor
  - b) Current through the conductor
  - c) Atmospheric condition
  - d) Ice deposition on conductor

- Q.4 Murray loop test works on the principle of
- a) Faraday's
  - b) Wheatstone bridge
  - c) Skin effect
  - d) None of the above
- Q.5 The String efficiency can be improved by :
- a) Using longer cross arms
  - b) Using a guard ring
  - c) Grading of the insulator
  - d) All of these
- Q.6 Diversity factor value is
- a) Always less than 1
  - b) Always greater than 1
  - c) Unity
  - d) None of these

### SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Expand ACSR.
- Q.8 String efficiency = \_\_\_\_\_.
- Q.9 The tendency of current to concentrate near the surface of the conductor is known as \_\_\_\_\_.
- Q.10 Define Power factor.
- Q.11 Define super heater.
- Q.12 HVDC stands for \_\_\_\_\_.

## SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the benefits of transmit electric power at high transmission voltage.
- Q.14 Explain the significance of load and diversity factor.
- Q.15 Describe the present scenario of energy in India.
- Q.16 What are the causes of low power factor?
- Q.17 Difference between feeders distributor & service mains.
- Q.18 Define Bundle conductors.
- Q.19 Write the components of a pole mounted substation and their function.
- Q.20 Explain the concept of regional and national grid.
- Q.21 Discuss the major equipment installed in a grid substations.
- Q.22 Enlist five merits and Demerits of Non-Conventional energy sources.

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the various types of supports used in transmission system.

- Q.24 Explain the construction and working of thermal power plant with a diagram.
- Q.25 Define power factor. Explain its significance in power system and method of improvement of power factor.